

Appendix S3. Parameter and configuration dictionary

ENA-ABM v0.5 main theoretical experiment

This appendix defines the configuration fields used to generate, execute, and analyze the version 0.5 experiment. It complements Appendix S1, which describes the model using the ODD protocol, and Appendix S2, which records the preregistered experimental design. The values are diagnostic theoretical inputs rather than empirically calibrated estimates.

The appendix distinguishes three configuration layers: (i) model fields accepted by one simulation run; (ii) design fields used to construct the 96 registered profiles and the relative-cost search; and (iii) analysis fields governing replication, bootstrap estimation, robustness variants, and interpretation stop rules. The machine-readable files supplied with this appendix are authoritative for exact execution.

S3.1 Machine-readable artifact set and precedence

Artifact	Contents	Use
main_study_v0.5.json	Experiment seed, replication counts, cost grid, main parameter ranges, fixed settings, robustness overrides, and stop-rule values	Primary machine-readable configuration for the v0.5 experiment
registered_profiles_v0.5.csv	One row for each profile P001–P096, including all nine varied factors and network type	Exact profile registry; use this file instead of regenerating the Latin hypercube
PREREGISTERED_DESIGN_v0.5.md	Frozen design rationale, estimands, analysis plan, and interpretation rules	Narrative authority for the preregistered design; source for Appendix S2
Version 0.4 source code	Validated ABM implementation, configuration schema, process rules, and output calculations	Executable authority for model behavior; source for Appendix S1
PROTOCOL_CORRECTION_v0.5.md	Timing, rationale, and limits of the separately executed fixed-cost comparison at $r = 1.20$	Authority for the documented protocol correction

If narrative rounding differs from a machine-readable value, the JSON and CSV values take precedence. The profile registry must not be reconstructed from the stated ranges because its deterministic row assignments are part of the frozen design.

S3.2 Value categories and notation

Category	Meaning
Profile-varying	A value recorded separately for each of the 96 registered profiles
Main fixed	A value held constant across the main profile experiment
Strategy-specific	A field whose value or use differs between direct and referral runs
Derived	A value calculated from one or more stored fields before or after a run
Robustness override	A prespecified replacement applied only in a named robustness context
Analysis control	A value governing replication, estimation, classification, or stopping rather than agent behavior

Probabilities and shares are dimensionless and lie in $[0, 1]$. Costs are expressed in normalized direct-attempt cost units, ticks are discrete simulation periods, and counts are nonnegative integers unless stated otherwise.

S3.3 Registered profile fields

The complete values appear in `registered_profiles_v0.5.csv`. Continuous profile factors were assigned by a deterministic stratified Latin-hypercube design. Budget and seed-adopter values were mapped to valid integers. Network type is balanced at 32 profiles per structure.

Field	Symbol or label	Type / unit	Registered domain	Definition
profile	Profile ID	String	P001–P096	Stable identifier joining design, simulation, and result records
market_sociality	λ	Probability weight	0.00–1.00	Market-level tendency for consumer evaluation to rely on social rather than individual information
fit_mean	μ_q	Share	0.35–0.80	Mean of the truncated-normal product–consumer fit distribution
status_mean	μ_s	Share	0.20–0.80	Target mean of the initial status-quo satisfaction beta distribution
uncertainty_mean	μ_u	Share	0.20–0.85	Target mean of the initial uncertainty beta distribution
beta_0	β_0	Logit coefficient	–1.50–0.50	Intercept in adoption utility
beta_individual	β_{ind}	Logit coefficient	1.00–6.00	Weight on individual appeal in adoption utility
beta_social	β_{soc}	Logit coefficient	0.00–6.00	Weight on local social information in adoption utility
budget_limit	B	Direct-cost units	8–60	Equal budget allocated separately to each strategy run
seed_adopters	n_0	Consumers	1–8	Consumers initialized as adopters before entrepreneurial actions begin
network_type	Network structure	Categorical	ring, small_world, preferential_attachment	Rule used to generate the undirected consumer network

S3.4 Main fixed model settings

S3.4.1 Population, network, and schedule

Field	Type / unit	Main value	Definition
n_consumers	Consumers	100	Number of consumer agents in each main-study run
mean_degree	Edges per consumer	6	Target degree for comparable ring and small-world networks and structural scale for preferential attachment

Field	Type / unit	Main value	Definition
rewire_probability	Probability	0.10	Watts–Strogatz edge-rewiring probability; relevant to the small-world structure
horizon	Ticks	30	Maximum number of simulation ticks
actions_per_tick	Initiated actions	2	Entrepreneurial action capacity per tick, subject to feasibility and remaining budget
allow_repeated_direct_contact	Boolean	true	Allows direct outreach to return to nonadopters after the uncontacted candidate pool is exhausted

For noncustom generated networks, the implementation requires `n_consumers` to be at least 2 and `mean_degree` to be even, at least 2, and smaller than `n_consumers`.

S3.4.2 Consumer heterogeneity and information

Field	Symbol	Type / unit	Main value	Definition
fit_sd	σ_q	Share	0.15	Standard deviation of product fit before truncation to $[0, 1]$
minimum_uncertainty	u_{min}	Share	0.05	Lower bound for uncertainty after information updates
delta_information	δ	Share	0.25	Magnitude of the uncertainty reduction generated by effective exposure, scaled by source credibility
direct_source_credibility	κ_D	Share	0.60	Credibility attached to information arriving through direct outreach
referral_source_credibility	κ_R	Share	0.60	Credibility attached to information arriving through referral outreach

Product fit is sampled from a normal distribution with profile mean `fit_mean` and standard deviation `fit_sd`, then bounded to $[0, 1]$. Initial status-quo satisfaction and uncertainty are beta-distributed. For a registered design mean m , the profile builder uses concentration 10 and sets

$$\alpha = 10m, \beta = 10(1 - m). \quad (\text{S3.1})$$

Thus the requested mean is preserved while dispersion is held comparable across profiles. Values passed to this mapping are bounded internally to $[0.01, 0.99]$; all registered means already lie inside that interval. Seed adopters are selected uniformly without replacement and do not count as new adoptions in the primary outcome.

S3.4.3 Adoption coefficients and dynamic classifications

Field	Type / unit	Main value or source	Definition
beta_0	Logit coefficient	Profile-varying	Baseline adoption propensity

Field	Type / unit	Main value or source	Definition
beta_individual	Logit coefficient	Profile-varying	Strength of product fit relative to status-quo satisfaction
beta_social	Logit coefficient	Profile-varying	Strength of the adopted-neighbor signal
market_sociality	Share	Profile-varying	Scales reliance on social information as uncertainty changes
takeoff_share	Share of nonseed market	0.10	Threshold used to classify takeoff in the primary analysis
quiet_period	Ticks	5	No-new-adoption window used in the primary extinction definition

These fields parameterize the common evaluation and adoption mechanism described in Appendix S1. Strategy does not enter the adoption equation directly; it affects which consumers obtain effective exposure and how much budget is consumed in producing it.

S3.5 Access-process configuration

Field	Route	Type	Main value	Definition
direct_exposure_probability	Direct	Probability	0.512	Conditional probability that an initiated direct contact becomes an effective exposure
referral_acceptance_probability	Referral	Probability	0.80	Probability that an available adopted source accepts the request
referral_neighbor_availability_probability	Referral	Probability	0.80	Conditional probability that the accepted source has a locally eligible nonadopter
referral_introduction_probability	Referral	Probability	0.80	Conditional probability that an eligible referral becomes an effective introduction

The neutral referral pipeline has conditional completion probability

$$p_R = 0.80 \times 0.80 \times 0.80 = 0.512, (S3.2)$$

which equals the main direct effective-exposure probability. This equality controls the conditional completion probability only. Referral outreach still requires an adopted source and a locally eligible neighbor; direct outreach can select from the wider eligible market. A referral opportunity with no adopted source is recorded as unavailable, does not initiate an attempt, and consumes no budget.

S3.6 Resource allocation and cost accounting

Field	Type / unit	Main setting	Definition
fairness_criterion	Categorical	equal_budget	Both strategies receive the same profile-specific allocated budget
budget_limit	Direct-cost units	Profile-varying, 8–60	Maximum expenditure permitted in one strategy run

Field	Type / unit	Main setting	Definition
direct_action_cost	Cost units	1.00	Normalized expected cost of one initiated direct attempt
referral_action_cost	Cost units	r	Expected cost of one initiated referral attempt
cost_accounting	Categorical	upfront_expected	Full expected attempt cost is charged when a feasible attempt begins
Direct stage shares	Four shares	0.25 each	Diagnostic search, contact, coordination, and follow-up shares used only under staged accounting
Referral stage shares	Four shares	0.25 each	Diagnostic identification, request, coordination, and follow-up shares used only under staged accounting

The relative-cost parameter is

$$r = \frac{C_R}{C_D}, C_D = 1, (S3.3)$$

so $r = 1$ means equal attempt cost, $r > 1$ means that a referral attempt is more costly, and $r < 1$ means that it is less costly. Under `upfront_expected`, failure after initiation remains costly. Under the robustness value `staged`, only stages reached are charged. The 0.25 stage shares are diagnostic accounting assumptions, not observed business cost shares.

The configuration class also contains `action_budget` and `exposure_quota` for alternative comparison criteria. They are not the binding fairness resources in the v0.5 main experiment; `budget_limit` under `equal_budget` is the operative constraint.

S3.7 Relative-cost search and threshold controls

Field	Type	Value	Role
relative_cost.coarse_grid	Ordered numeric vector	0.25, 0.50, ..., 3.00	Initial values of r evaluated for every profile
relative_cost.refinement_tolerance	Cost-ratio increment	0.05	Spacing of intermediate evaluations inside every sign-change bracket
replicates	Paired replications per cell	400	Monte Carlo sample used for each profile–cost comparison
bootstrap_samples	Paired resamples	500	Within-profile resamples used to estimate threshold uncertainty
experiment_seed	Integer	2026072505	Master experiment namespace for deterministic profile execution

The profile-specific competitive threshold is the largest evaluated r for which the mean paired adoption difference is nonnegative. Profiles without a competitive region and profiles competitive through $r = 3.00$ remain absent and right-censored, respectively; the algorithm does not force a finite crossing.

S3.8 Randomization, pairing, and reproducibility fields

Field or stream	Scope	Definition
experiment_seed	Complete v0.5 experiment	Reproduces design execution and analysis when combined with the frozen code and artifacts
master_seed	Simulation namespace	Root value from which stable component seeds are derived for a configured run
replicate_id	One paired market realization	Identifies a replication within a profile and cost cell
network_seed	Component stream	Network generation
attribute_seed	Component stream	Consumer fit, status satisfaction, and uncertainty
seed_selection_seed	Component stream	Initial adopter selection
strategy_seed	Component stream	Strategy-level selection operations
exposure_process_seed	Component stream	Direct exposure and referral-stage outcomes
decision_seed	Component stream	Adoption draws

Direct and referral runs within a profile–replication pair share the initialized network, consumer attributes, and seed adopters. Component-specific deterministic streams prevent an unrelated random draw in one process from silently changing another process. Reproduction therefore requires the frozen profile row, replicate identifier, cost ratio, experiment configuration, source code, and seed construction—not merely the headline seed.

S3.9 Process-robustness override dictionary

Blank entries retain the neutral main-study value. Each regime is evaluated separately for ring, small-world, and preferential-attachment networks.

Regime key	Direct exposure	Acceptance	Eligible neighbor	Introduction
neutral	0.512	0.80	0.80	0.80
equal_high	0.729	0.90	0.90	0.90
direct_exposure_advantage	0.80	0.80	0.80	0.80
referral_exposure_advantage	0.512	0.90	0.90	0.90
acceptance_low	0.512	0.40	0.80	0.80
acceptance_high	0.512	1.00	0.80	0.80
eligible_neighbors_none	0.512	0.80	0.00	0.80
eligible_neighbors_abundant	0.512	0.80	1.00	0.80

The equal-high direct value is $0.729 = 0.90^3$. The process variants change access probabilities only and do not modify the common evaluation or adoption mechanism.

S3.10 Accounting-robustness contexts

Each context is run under upfront_expected and staged accounting while all unlisted values remain at the baseline context setting.

Context key	Override
baseline	None
sociality_low	market_sociality = 0.10
sociality_high	market_sociality = 0.90
budget_scarce	budget_limit = 8
budget_abundant	budget_limit = 60

Context key	Override
referrers_scarce	seed_adopters = 1
referrers_abundant	seed_adopters = 8
network_ring	network_type = ring
network_preferential_attachment	network_type = preferential_attachment

S3.11 Structural-robustness contexts

Context key	Override
baseline	None
network_size_small	n_consumers = 50
network_size_large	n_consumers = 200
mean_degree_low	mean_degree = 4
mean_degree_high	mean_degree = 10
rewiring_zero	rewire_probability = 0.00
rewiring_high	rewire_probability = 0.30
network_ring	network_type = ring
network_preferential_attachment	network_type = preferential_attachment

These contexts alter existing structural parameters only; they do not introduce new agents, decision rules, or diffusion mechanisms.

S3.12 Dynamic-definition robustness

Field	Values	Effect
relative_cost_values	1.00, 1.20	Cost settings at which alternative dynamic classifications are summarized
takeoff_shares	0.05, 0.10, 0.15, 0.20	Alternative shares of the nonseed market used to classify takeoff
quiet_periods	3, 5, 10 ticks	Alternative no-new-adoption windows used to classify extinction

These alternatives recalculate classifications from simulated histories; they do not change exposure, evaluation, or adoption events.

S3.13 Analysis stop-rule configuration

JSON field	Value	Operational meaning
multiple_crossing_share_max	0.10	Maximum tolerated share of main profiles with multiple competitive-region crossings before stopping substantive threshold interpretation
unstable_bootstrap_interval_width_max	0.50	Width used to flag unstable profile-level bootstrap threshold intervals
accounting_difference_material	0.25	Cost-ratio difference treated as materially important in the accounting comparison
budget_rounding_share_material	0.10	Share used to flag material budget-residual or rounding influence
minimum_interpretable_profile_share	0.50	Minimum share of profiles required to yield interpretable competitive regions
reproducibility_required	true	Requires an independent complete reproduction before accepting the analysis package

These fields are interpretation controls. They do not remove scientifically valid low-adoption, zero-adoption, unavailable-opportunity, extinction, absent-region, or right-censored runs.

S3.14 Derived quantities and outcome fields

Quantity or outcome	Definition
New adopters	Final adopters minus initial seed adopters; primary strategy outcome
Paired adoption difference	Referral new adopters minus direct new adopters for the same profile and replication
r^*	Largest evaluated relative cost with a nonnegative mean paired adoption difference, subject to profile classification
Unavailable referral opportunity	Scheduled referral action for which no adopted source is available; no attempt and no cost
Referral request accepted	Initiated referral attempt that passes the acceptance stage
Eligible neighbor	Accepted source with a locally eligible nonadopter after the availability process
Effective introduction	Eligible referral that completes the introduction stage and creates exposure
Effective direct exposure	Initiated direct attempt that passes its exposure process
Conversion after exposure	New adoption divided by effective exposures, reported separately from access generation
Budget remaining	Allocated budget minus charged strategy cost at termination
Takeoff	New adoption reaches the configured share of the nonseed population
Extinction	No takeoff, entrepreneurial action has ended, and no new adoption occurs during the configured quiet period
Cascade depth and reach	Referral-parent-based measures of how far adoption descendants extend from seed adopters

S3.15 Validation domains and implementation safeguards

The configuration validator requires probabilities and shares to remain in $[0, 1]$; positive action costs; nonnegative resource limits; at least one action per tick; a positive horizon and quiet period; positive beta-distribution shape parameters; valid strategy, network, fairness, and accounting labels; and direct and referral stage shares that each sum to one. The seed-adopter count must be nonnegative and smaller than the population.

Every run records a configuration and initialization signature. Paired strategy runs must have identical initialization signatures. Invalid probabilities, impossible resource settings, unknown fields, mismatched initialization, or malformed networks raise errors rather than being silently repaired during execution.

S3.16 Reproduction note and interpretation boundary

To reproduce the main experiment, use the frozen version 0.4 model implementation together with `main_study_v0.5.json` and the supplied `registered_profiles_v0.5.csv`. Do not sample new profile values from the reported ranges. The separately documented $r=1.20$ comparison completes a preregistered contrast but does not change the primary grid or threshold estimates.

The parameter dictionary defines a theoretical design space. Values such as 0.80 acceptance, 0.60 credibility, or the 8–60 budget range are controlled experimental inputs. They must not be interpreted as measured market averages, estimated causal effects, or recommended managerial constants. Empirical calibration and external validation are required before translating a simulated threshold into a firm-specific decision rule.